

Problem Set 7
Due: Apr. 12, 2021

1. Section 3.4 Problem 16.

Consider the following variant of Buffon's needle problem (Example 3. 11), which was investigated by Laplace. A needle of length l is dropped on a plane surface that is partitioned in rectangles by horizontal lines that are a apart and vertical lines that are b apart. Suppose that the needle's length l satisfies $l < a$ and $l < b$. What is the expected number of rectangle sides crossed by the needle? What is the probability that the needle will cross at least one side of some rectangle?

2. Section 3.5 Problem 20.

An absent-minded professor schedules two student appointments for the same time. The appointment durations are independent and exponentially distributed with mean thirty minutes. The first student arrives on time, but the second student arrives five minutes late. What is the expected value of the time between the arrival of the first student and the departure of the second student?

3. Section 3.6 Problem 34.

A defective coin minting machine produces coins whose probability of heads is a random variable P with PDF

$$f_P(p) = \begin{cases} pe^p & p \in [0, 1] \\ 0 & \text{otherwise} \end{cases}$$

A coin produced by this machine is selected and tossed repeatedly, with successive tosses assumed independent.

- (a) Find the probability that a coin toss results in heads.
- (b) Given that a coin toss resulted in heads, find the conditional PDF of P .
- (c) Given that the first coin toss resulted in heads, find the conditional probability of heads on the next toss.

4. Let Δ denote the event that you can form a triangle with three given parts of a rod R .

- (a) R is broken in two uniformly at random, the longer part is broken in two uniformly at random. Show that $P(\Delta) = \log(4/e)$.
- (b) R is broken in two uniformly at random, a randomly chosen part is broken into two equal parts. Show that $P(\Delta) = \frac{1}{2}$.
- (c) In case (b) show that, given Δ , the triangle is obtuse with probability $3 - 2\sqrt{2}$.